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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 03/11/2020 |
| **PI Name:** | Gupta, Anjali Ph.D. | | |
| **Project Title:** | An 'Omics' Approach to Identify the Differential Effects of Altered Gravity on Skeletal vs. Cardiac Muscle | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Invertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | kocorr@sbpdiscovery.org | **Fax:** | FY |
| **PI Organization Type:** | NON-PROFIT | **Phone:** | 858-646-3100 |
| **Organization Name:** | Sanford Burnham Prebys Medical Discovery Institute | | |
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| **City:** | La Jolla | **State:** | CA |
| **Zip Code:** | 92037 | **Congressional District:** | 49 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research |
| **Start Date:** | 08/01/2019 | **End Date:** | 12/11/2019 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/11/2019 (original end date was 7/31/2021) due to postdoctoral fellow's relocation (Ed., 3/12/2020) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Ocorr, Karen Ph.D. ( MENTOR: Sanford Burnham Prebys Medical Discovery Institute )<br>Bhattacharya, Sharmila Ph.D. ( NASA Ames Research Center ) | | |
| **Grant/Contract No.:** | 80NSSC19K1295 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | POSTDOCTORAL FELLOWSHIP<br>BACKGROUND: While the effects of gravity on cardiac and skeletal muscle are well established at the physiological level, the genes and molecular pathways underlying these gross changes are largely unknown. Furthermore, the dose-dependent relationship of gravity as a continuum on biological systems is still unknown. In order to develop effective countermeasures for gravity induced cardiomyopathy and skeletal muscle atrophy, it is important to identify the genes and respective molecular signaling pathways that are affected by altered gravity. The fruit fly, Drosophila melanogaster, shares 75% homology with human diseased genes. With only four chromosomes, non-redundant evolutionarily conserved genes and pathways, and numerous established techniques for conducting functional experiments, the fruit fly provides a meaningful opportunity to examine the fundamental effects of gravity as an environmental variable and to identify potential targets for treatment of spaceflight induced dysfunction. In addition, the short life span of Drosophila will allow us to identify the effects of extended exposure to altered gravity with age, something that would take decades to identify in humans.<br><br>OBJECTIVES: The primary objective of this investigation is to systematically study the effects of a GRAVITY CONTINUUM (0g, 1g, and 3g) on striated muscle such as cardiac and skeletal muscle by leveraging our previous spaceflight data in combination with new ground based hyper-gravity studies. Our goal is to: (1) to conduct functional cardiac analysis in vivo using a simple, genetically tractable animal model, to explore how altered gravity affects cardiac and skeletal muscle function, structure, and gene transcription, which is likely to identify new molecular pathways underlying gravity-induced dysfunction, (2) to assess the effects of gravity on gene and protein expression in cardiac and skeletal muscle, and potentially identify molecular pathways differentially regulated in response to altered gravity, (3) to study striated muscle plasticity in a mutant animal model by assessing structure, function, and gene and protein expression following exposure to altered gravity and comparing the results to wildtype animals.<br><br>METHODS: Data have already been collected from flies aboard the International Space Station (ISS) (<1g) compared to ground controls (1g). An industrial grade centrifuge will be used to expose fruit flies to 3g gravity conditions for comparison with a new set of 1g ground controls. Tissue from the heart and thorax will be collected for transcriptomic (via RNAseq) and proteomic (via mass spectrometery) analyses. The resulting data will be analyzed using publicly available bioinformatics tools and standard practices.<br><br>SIGNIFICANCE: The novelty and significance of this proposal lies in the ability to determine the effects of gravity on multiple organ systems and formulate a systems-level view of the effects of gravity on the organism. This systemic investigation of gravity as a "dosage" in a genetically simple organism with evolutionary conservation allows us to gain novel insights into the molecular and mechanistic etiology of gravitational effects. These analyses will potentially confirm common genetic pathways between animal models and/or identify new and unique pathways for future interrogations. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Analyzed data from CRS-3 and CRS-11 missions. Data analyzed included heart function analysis, negative geotaxis, and fluorescently stained and optically sectioned fly hearts. Wildtype hearts were compared against D45 mutants (muscle dysfunction).<br>Collected tissues and extracted RNA from fly hearts for RNAseq analysis.<br><br>[Ed. note, March 2020: project ended early due to relocation of postdoctoral fellow] |
| **Bibliography Type:** | Description: (Last Updated: ) |